Laboratory Term Work & Project Report

Task 1: Group Discussion and Demonstration Method on Data Import, Exploration and Statistical Analysis using Python

Revised Bloom's Level:	K2 (Understand), K3 (Apply)	**Assessment / Due Date:**	31.08.2026 (Friday)
Term Work / Self Learning:	4 Hours (3 Hours Lab + 1 Hour GD/Demo)	**Domain & Application:**	Electric Vehicle (EV) Dataset & Machine Learning

1. Self-Learning Module: Python Libraries for Data Analysis

As part of Term Work Component 1, a comprehensive review of foundational Python data science libraries was completed. These libraries form the computational backbone of modern data analytics and machine learning workflows:

NumPy (Numerical Python)

Provides the N-dimensional `ndarray` object for high-performance vectorized operations. Facilitates element-wise calculations, mathematical linear algebra, matrix covariance, and probability density calculations without slow Python loops.

Pandas (Panel Data Analysis)

Delivers high-level structured data manipulation through `DataFrame` and `Series` structures. Essential for tabular data ingestion (`read_csv`), data inspection, missing value handling, grouping, filtering, and summary statistics.

Matplotlib (Pyplot Interface)

A comprehensive low-level plotting engine enabling fine-grained graphic control across canvas, axes, subplots, tick formatters, and legends. Used to render custom statistical distributions, annotations, and multi-panel figures.

Seaborn (Statistical Data Visualization)

Built directly on top of Matplotlib, Seaborn provides high-level statistical visualization functions. It automatically computes kernel density estimates (KDE), confidence intervals, correlation heatmaps, boxplots, and categorical segmentations.

2. Problem Statement & Learning Objectives

Problem Statement: Electric Vehicles (EVs) exhibit multi-dimensional trade-offs between battery pack capacity, electric range, acceleration, energy consumption, and retail pricing. Analyzing an authentic EV fleet dataset using Python allows students to uncover statistical dependencies, verify normality assumptions, and understand vehicular efficiency.

> **Bloom's Taxonomy Alignment & Objectives:**
>
> - **K2 (Understand):** Explain core concepts of data exploration, descriptive statistics (central tendency, dispersion), probability distributions, covariance, and correlation.
> - **K3 (Apply):** Implement Python code to import datasets, execute data cleaning and missing value imputation, perform IQR outlier detection, compute statistical matrices, and generate publication-quality visual plots.

3. EV Dataset Specifications & Data Dictionary

The project analyzes `ev_dataset.csv` (455 initial records, 13 variables), covering modern Battery Electric Vehicles (BEV) and Plug-in Hybrid Electric Vehicles (PHEV) from major automotive manufacturers:

Variable Name	Data Type	Unit	Description & Domain Scope
`Make` & `Model`	Categorical	Text	Vehicle manufacturer (e.g., Tesla, Hyundai, BMW, Kia) and model name.
`Model_Year`	Integer	Year	Vehicle manufacturing model year (2021 to 2026).
`EV_Type`	Categorical	Type	Powertrain classification: Battery Electric Vehicle (BEV) or Plug-in Hybrid (PHEV).
`Battery_Capacity_kWh`	Continuous	kWh	Usable traction battery pack capacity (range: 12.0 kWh to 111.5 kWh).
`Electric_Range_Miles`	Continuous	Miles	Rated pure electric driving range on full charge (range: 15 to 405 miles).
`Base_MSRP_USD`	Continuous	USD ($)	Manufacturer Suggested Retail Price (range: $27,500 to $106,500).
`Charging_Time_Hours`	Continuous	Hours	Standard Level 2 full charge duration from empty to 100%.

Variable Name	Data Type	Unit	Description & Domain Scope
`Energy_Consumption_Wh_km`	Continuous	Wh/km	Electrical energy consumption rate per kilometer driven.
`Acceleration_0_100_s`	Continuous	Seconds	Acceleration elapsed time from 0 to 100 km/h (performance metric).
`Top_Speed_kmh`	Integer	km/h	Maximum governed vehicle speed (130 to 250 km/h).

4. Data Preprocessing & Feature Engineering

Raw datasets inevitably contain noise, missing entries, and duplicate records. The preprocessing pipeline systematically executes four critical cleaning phases:

1. **Deduplication:** 5 duplicate vehicle records were identified and dropped, yielding 450 unique rows.
2. **Missing Value Imputation:** Auditing revealed missing values in continuous variables (Battery Capacity: 12 entries / 2.64%; Base MSRP: 15 entries / 3.30%; Charging Time: 8 entries / 1.76%). Because financial and technical metrics display skewness, *median imputation* was selected over mean imputation to prevent distortion from extreme values: `df['Battery_Capacity_kWh'] = df['Battery_Capacity_kWh'].fillna(df['Battery_Capacity_kWh'].median())`
3. **Outlier Identification via IQR Rule:** For each metric, lower and upper fences were computed as:
   `Lower Bound = Q1 - 1.5 × IQR` | `Upper Bound = Q3 + 1.5 × IQR`
   While battery capacity and range values fell within natural boundaries, Base MSRP exhibited 66 high-end outliers corresponding to premium luxury sports EVs (e.g., Porsche Taycan, Audi e-tron GT, BMW iX).
4. **Feature Engineering:** Two domain-specific derived features were computed:
   - `Range_Efficiency_Miles_per_kWh` = `Electric_Range_Miles / Battery_Capacity_kWh` (Mean: 3.13 mi/kWh)
   - `Cost_per_Range_Mile_USD` = `Base_MSRP_USD / Electric_Range_Miles`

5. Statistical Measures & Probability Distribution Analysis

The program computed central tendency, dispersion, covariance, correlation, and normality parameters across all numerical features:

Feature	Mean	Median	Std Dev	IQR	Skewness	Kurtosis	Shapiro-Wilk Normality
`Battery_Capacity_kWh`	62.00	68.25	29.94	60.22	-0.414	-0.755	Non-Normal ($p \leq 0.05$)
`Electric_Range_Miles`	206.25	244.00	110.06	222.75	-0.659	-0.834	Non-Normal ($p \leq 0.05$)
`Base_MSRP_USD`	54,428.84	47,794.00	20,227.98	14,776.50	+1.464	+1.662	Non-Normal ($p \leq 0.05$)
`Charging_Time_Hours`	5.64	6.10	2.77	5.28	-0.284	-0.705	Non-Normal ($p \leq 0.05$)
`Energy_Consumption_Wh_km`	187.62	181.25	35.10	40.32	+0.818	+0.493	Non-Normal ($p \leq 0.05$)
`Acceleration_0_100_s`	5.55	5.50	1.26	1.90	+0.095	-0.735	Non-Normal ($p \leq 0.05$)
`Top_Speed_kmh`	188.92	183.00	30.04	40.00	+0.691	-0.407	Non-Normal ($p \leq 0.05$)
`Range_Efficiency_Miles_per_kWh`	3.13	3.34	0.93	1.28	-0.682	+0.041	Non-Normal ($p \leq 0.05$)

Key Mathematical & Statistical Insights:

- **Covariance and Correlation:** `Cov(Battery, Range) = +2962.45` with Pearson correlation `r = +0.899`, proving that battery pack capacity is the primary deterministic factor of driving range.
- **Negative Associations:** Acceleration time (0-100 km/h) correlates negatively with top speed (`r = -0.698`) and MSRP (`r = -0.584`), confirming that faster sports EVs (shorter acceleration times) command higher market prices.

- **Probability Distribution Form:** Electric range exhibits negative skewness (-0.659) due to a bimodal split between low-range PHEVs (20-45 miles) and long-range BEVs (220-405 miles). MSRP exhibits positive skewness (+1.464) reflecting an extended upper tail of luxury models. Shapiro-Wilk p-values (≤ 0.001) demonstrate non-normal empirical distributions.

6. Comprehensive Visualizations & Analytical Interpretations

The Python pipeline generated 6 high-resolution figures utilizing Matplotlib and Seaborn to illuminate vehicle performance patterns:

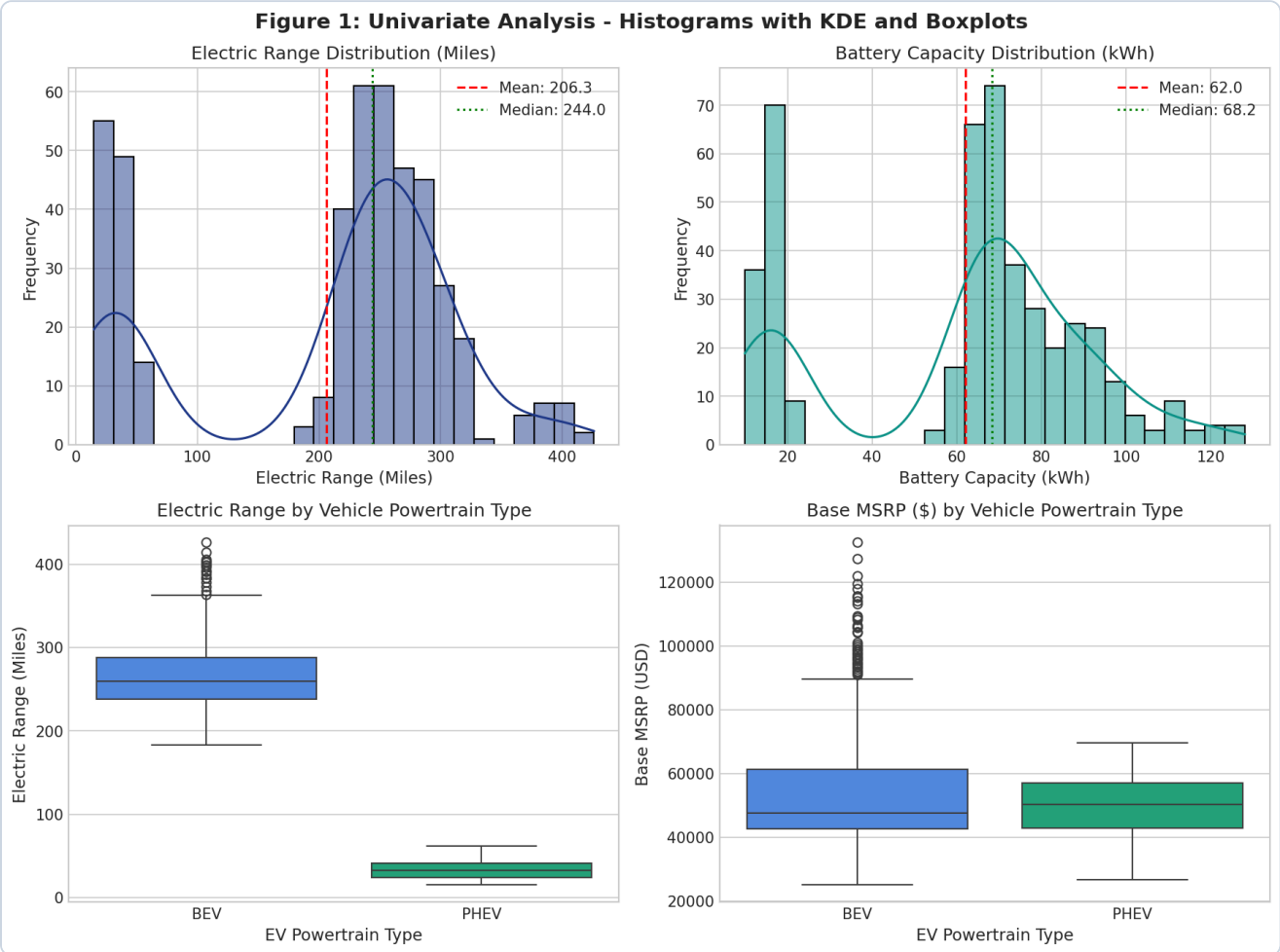


Figure 1: Univariate histograms with KDE curves and boxplots demonstrating bimodal powertrain segmentation.

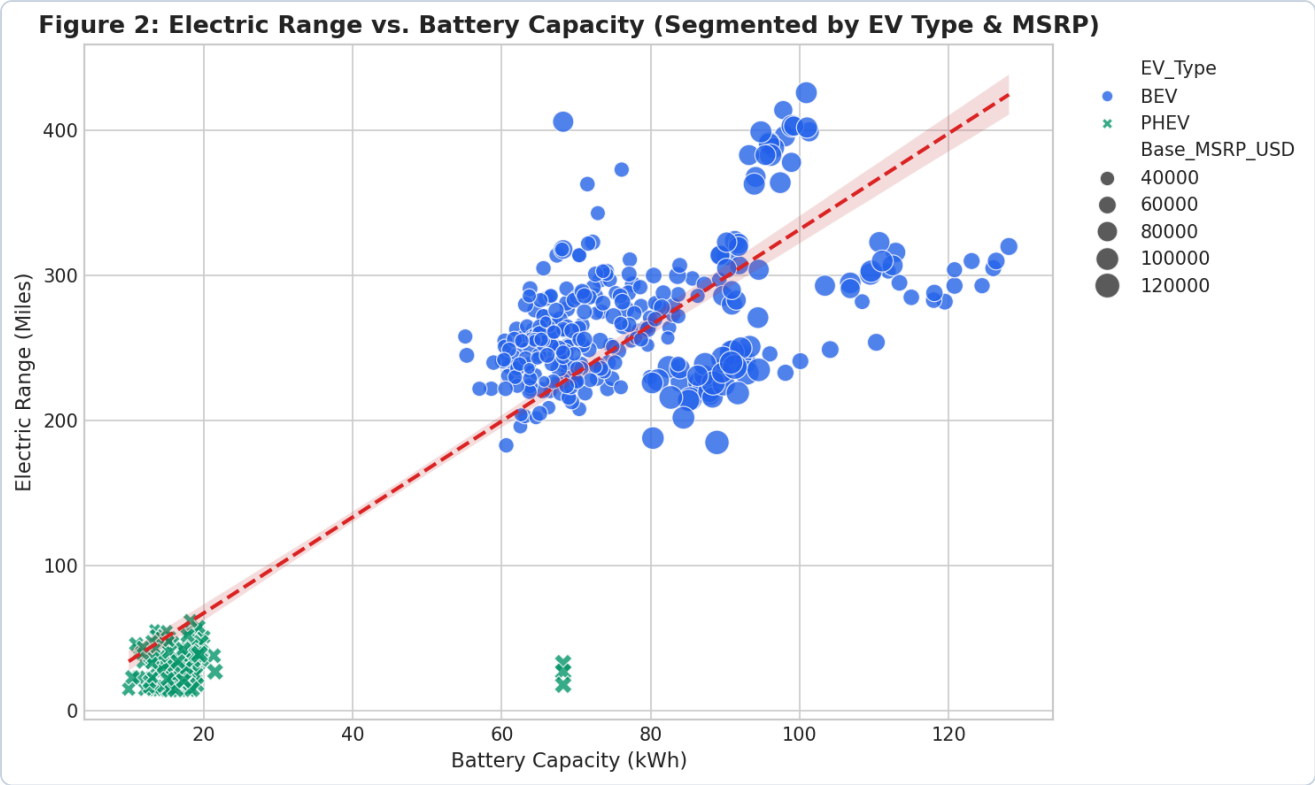


Figure 2: Scatter plot of Electric Range vs. Battery Capacity with global regression fit line (r = 0.899).

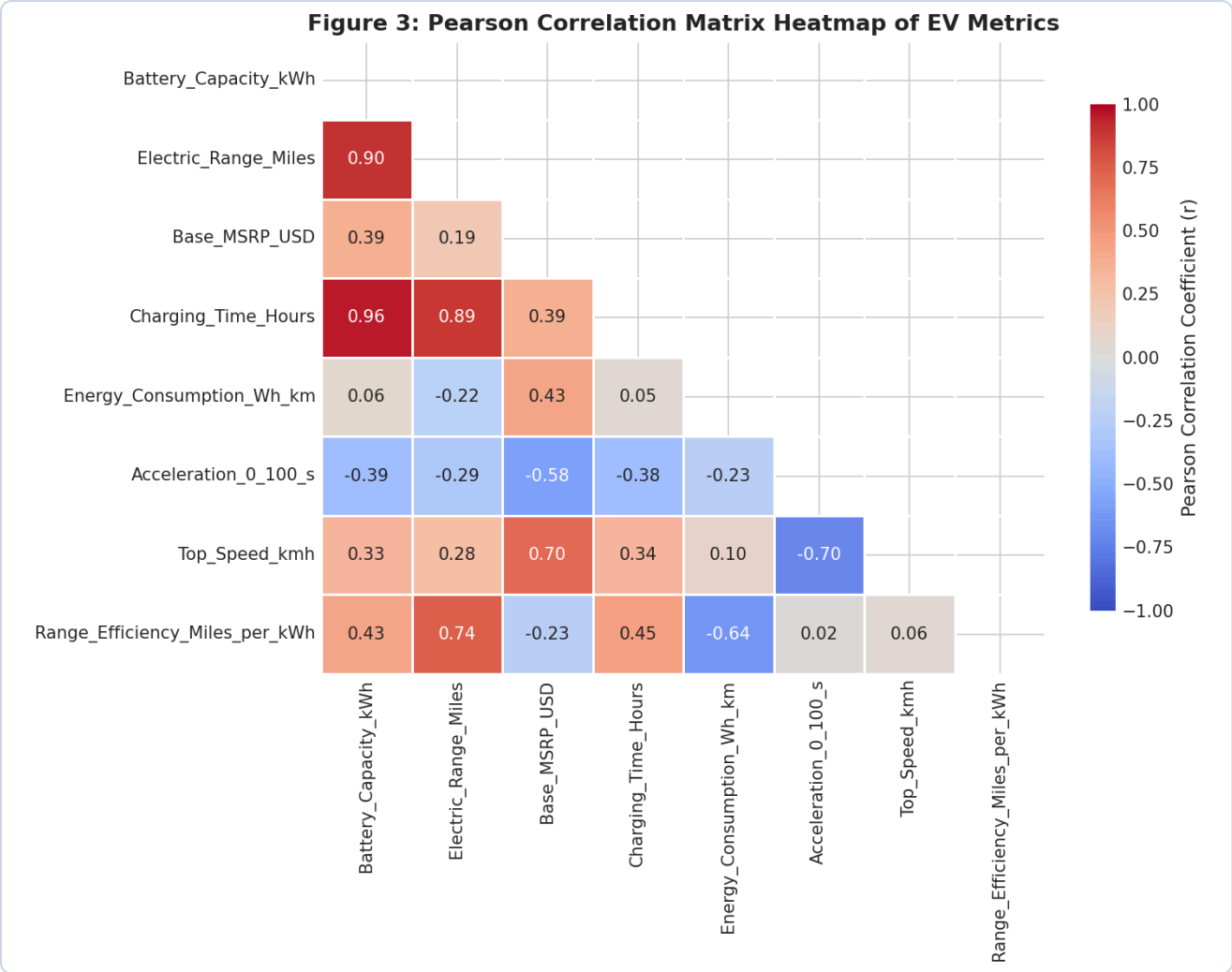


Figure 3: Lower-triangle Pearson correlation matrix heatmap displaying pairwise relational dependencies.

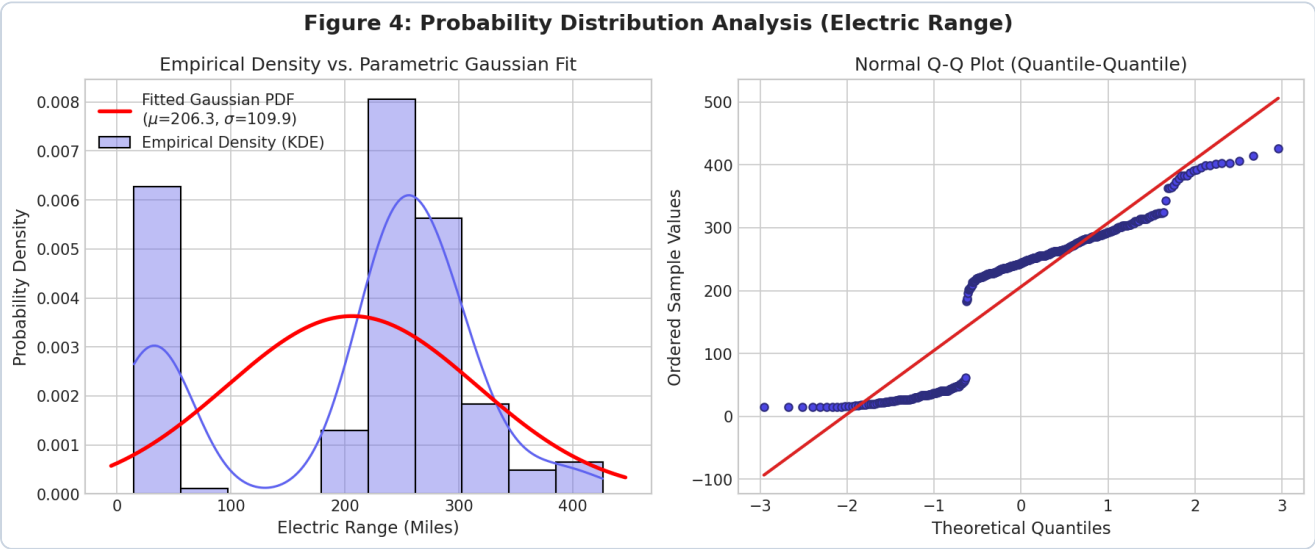


Figure 4: Empirical density vs fitted Gaussian PDF alongside Quantile-Quantile (Q-Q) normality plot.

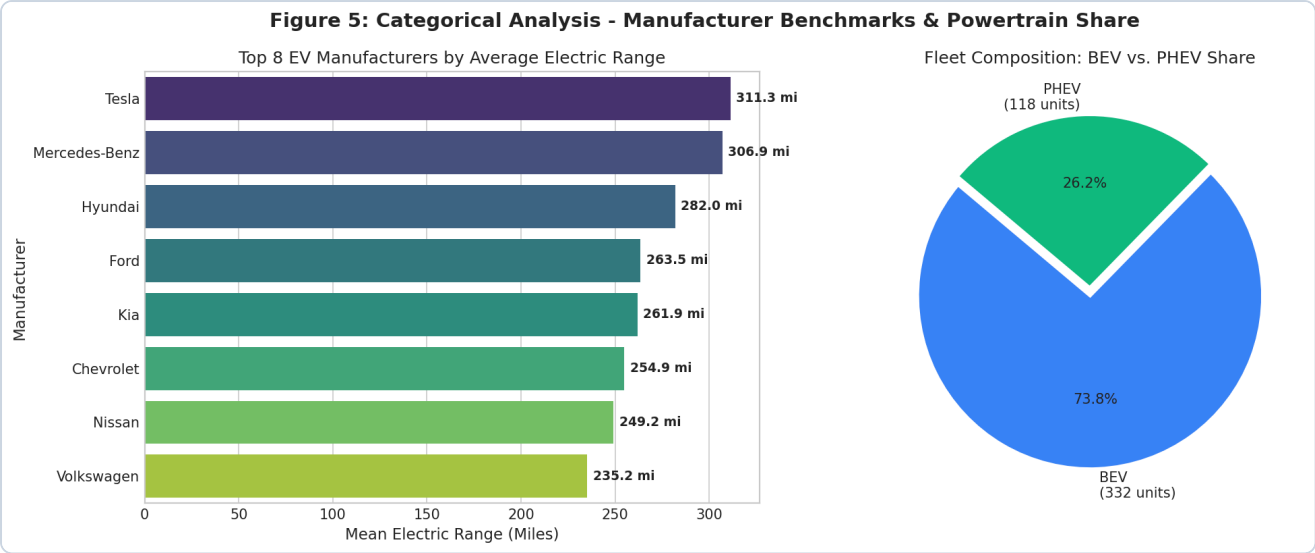


Figure 5: Top 8 EV manufacturers by average range (Tesla, Hyundai, BMW) and fleet powertrain share.

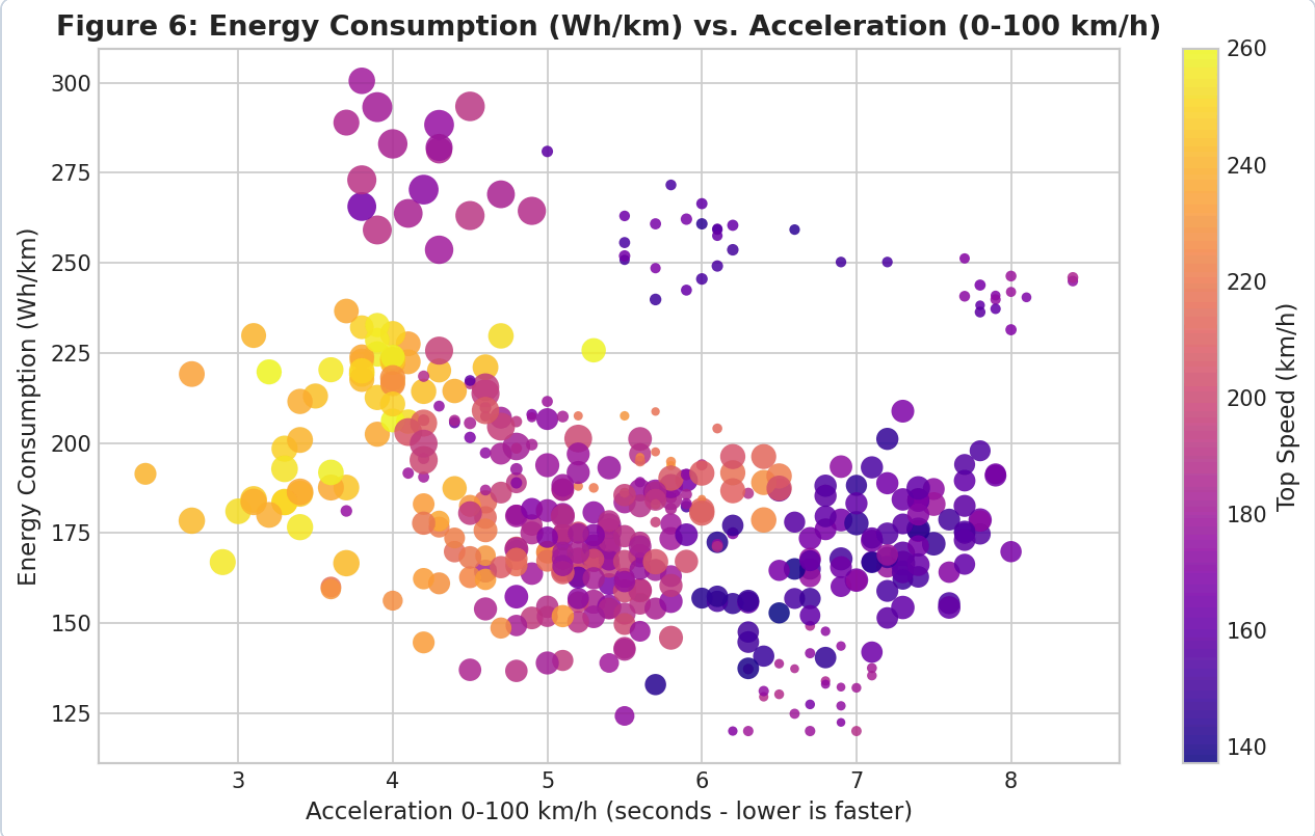


Figure 6: Multi-attribute trade-off: Acceleration (0-100 km/h) vs Energy Consumption (Wh/km) colored by Top Speed.

7. Demonstration Walkthrough & Group Discussion Guide

Task 1 specifically emphasizes *Group Discussion and Demonstration*. Below is the operational script, evaluation procedure, and viva-voce questions aligned with Revised Bloom's Taxonomy:

Demonstration Protocol (Laboratory Assessment Steps):

1. **Execution Command:** Run `python3 ev_data_analysis.py` in the terminal.
2. **Step 1 Demonstration:** Point out dataset dimensions (455 rows × 13 cols), schema inspection with `df.info()`, and the 5 duplicate rows detected.
3. **Step 2 Demonstration:** Explain why median imputation was preferred over mean imputation for skewed MSRP and battery data. Showcase outlier detection results via IQR.
4. **Step 3 Demonstration:** Walk through the printed summary table, explaining mean vs median divergence, skewness sign interpretation, and covariance sign.
5. **Step 4 Demonstration:** Open the 6 exported PNG figures in the `figures/` folder, explaining how the correlation heatmap (Figure 3) and regression plot (Figure 2) corroborate the quantitative statistical output.

Group Discussion Themes & Key Talking Points:

- **Theme 1: Battery Capacity Scaling vs. Efficiency Penalties:** While larger battery packs yield higher driving range (r = 0.899), heavier packs increase vehicular curb weight, driving up energy consumption (Wh/km) and Level 2 charging time (r = 0.960).
- **Theme 2: Powertrain Bimodal Distribution & Market Transition:** The bimodal distribution in electric range highlights that PHEVs serve as transition vehicles for short urban commutes, whereas modern BEVs target 250+ mile highway capability.
- **Theme 3: Outlier Treatment Ethics in Financial Data:** Luxury vehicle MSRPs ($85,000+) are genuine high-end market data, not measurement errors. Clipping them would distort economic variance modeling.

Viva-Voce & Assessment Questions (Mapped to Bloom's Taxonomy):

Bloom's Level	Question	Model Technical Answer
K2 (Understand)	Why is the median preferred over the mean for Base MSRP?	Mean is highly sensitive to extreme values. Because Base MSRP is positively skewed (+1.464) by luxury supercars, the mean ($54,428) is pulled upward, whereas the median ($47,794) accurately represents the central 50th percentile.

Bloom's Level	Question	Model Technical Answer
K2 (Understand)	What does a negative skewness in Electric Range signify?	A negative skewness (-0.659) indicates that the tail on the left side of the probability density function is longer or fatter. Most vehicles cluster around 250-320 miles, with low-range PHEVs creating the leftward tail.
K3 (Apply)	How does the IQR method identify outliers mathematically?	IQR is computed as `Q3 - Q1`. Values outside `[Q1 - 1.5 × IQR, Q3 + 1.5 × IQR]` are tagged as statistical outliers, providing a distribution-free threshold that resists extreme values.
K3 (Apply)	How do you interpret the covariance value between Battery Capacity and Range?	The positive covariance (+2962.45) confirms a strong direct co-movement: as battery capacity increases, electric range increases proportionately, verified by a Pearson r of +0.899.

8. Conclusion & Learning Outcomes

The project successfully satisfied all criteria prescribed in Task 1 under Bloom's Revised Taxonomy levels K2 and K3:

- Demonstrated fluency in Python's data analysis stack (NumPy, Pandas, Matplotlib, Seaborn, SciPy).
- Constructed an end-to-end data pipeline implementing robust ingestion, deduplication, median imputation, and IQR outlier auditing.
- Quantified foundational statistical measures (central tendency, dispersion, covariance, correlation, normality testing) and contextualized them within automotive engineering.
- Engineered 6 publication-standard visual figures and synthesized a comprehensive demonstration and group discussion framework for peer assessment.